II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 13

Kidney & UTUC

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · **P2** = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 136	Benign Renal Tumours
Ch 137	Malignant Renal Tumours
Ch 138	Urothelial Tumours of the Upper Urinary Tract and Ureter
Ch 139	Surgical Management of Upper Urinary Tract Urothelial Tumours
Ch 141	Contemporary Open Surgery of the Kidney
Ch 142	Laparoscopic and Robotic Surgery of the Kidney
Ch 143	Focal Therapy for Renal Tumours
Ch 144	Treatment of Advanced Renal Cell Carcinoma
★	Putting it together

136 Benign Renal Tumours

1 · THE BENIGN SMALL RENAL MASS

Many resected "cancers" are benign.

- Up to **~25%** of surgically treated small renal masses (SRMs) are benign.
- Only the **simple cyst** and **fat-containing angiomyolipoma (AML)** are imaging diagnoses; size, sex and age are unreliable.
- Management follows the risk of renal cell carcinoma (RCC); the SRM algorithm is in Ch 137.

2 · RENAL CYSTS AND THE BOSNIAK CLASSIFICATION

- A **simple cyst** (any size, homogeneous, **≤20 Hounsfield units (HU)**) needs no further imaging. Otherwise use Bosniak.

Class	Features (v2019)	Malignancy	Management (Canadian Urological Association, CUA 2023)
I	Thin (**≤2 mm**) smooth wall (may enhance); simple fluid; no septa or calcification	Benign	No follow-up
II	Thin wall plus (a) **≤3** thin enhancing septa, any calcification; or (b) homogeneous marked T2 hyperintensity; or (c) marked T1 hyperintensity	Highly likely benign	No follow-up
IIF	Minimally thickened (**3 mm**) enhancing wall or septum, or **≥4** thin septa; or heterogeneous T1 hyperintensity on fat-saturated MRI	Likely benign	Active surveillance up to **5 yr**

Class	Features (v2019)	Malignancy	Management (Canadian Urological Association, CUA 2023)
III	Enhancing thick (**≥4 mm**) or irregular (≤3 mm obtuse-margined protrusion) wall or septum	**~50%**	≤2 cm: surveillance preferred; 2–4 cm: surveillance or excision (shared decision); >4 cm: excision, partial (PN) over radical nephrectomy (RN) when feasible
IV	Enhancing nodule: ≥4 mm obtuse-margined, or any-size acute-margined protrusion	**~90%**	

KEY POINT

Ladder: wall ≤2 mm, 3 mm, ≥4 mm, irregular, then nodule. Traps: any calcification fits II, a class I wall may enhance, and growth of I/II is not malignancy.

KEY POINT

Why surveil small III/IV? Cystic RCC is indolent and mostly fluid, so size predicts poorly (proposed **pT1c**) and the number needed to treat is very high.

- European Association of Urology (EAU): treat Bosniak IV as a solid mass, and biopsy a solid component only if it changes management.

Diffuse cystic disease

P2 *ADPKD (Adult polycystic kidney disease); Acquired renal cystic disease — genetics, extrarenal cysts, US criteria, tolvaptan, cyst complications; ACKD prevalence, ADPKD vs ACKD*

Not in the paper answer:

- Acquired cystic kidney disease (ACKD) affects **80%** by 10 yr of dialysis, with RCC risk **≥10×** (multifocal, bilateral, often papillary).
- PKD1 relatives aged 15–29 need **≥3 cysts** for diagnosis; a normal scan excludes disease only from age 40.

3 · ONCOCYTOMA

Biopsy cannot reliably separate it from chromophobe RCC.

- It is the **commonest benign enhancing renal mass**, up to half of benign SRMs <4 cm.
- **Birt–Hogg–Dubé** (folliculin, chr 17) gives fibrofolliculomas, lung cysts with pneumothorax, oncocytoma and chromophobe RCC (chRCC).
- A **stellate scar** is non-specific, whereas **segmental enhancement inversion** favours oncocytoma.

	Oncocytoma	Chromophobe RCC
Cytokeratin 7 (CK7)	Rare, single-cell; KIT positive	Diffuse positive
Cytogenetics	Normal, −1, −Y, 11q13 rearrangement	Extensive abnormalities

KEY POINT

Why can a core not clear oncocytoma? chRCC has oncocytic areas, so some biopsy "oncocytomas" are RCC. EAU: no definitive core diagnosis.

4 · ANGIOMYOLIPOMA

AML can bleed catastrophically, so treatment follows bleeding risk.

P3 P2 *Management of AML; Renal angiomyolipoma — PEComa, TSC, mTOR, fat on CT, epithelioid variant, triggers, SAE, NSS, everolimus*

Not in the paper answer:

- An observed AML ≥4 cm needs CT angiography to look for an aneurysm.
- EAU (sporadic AML): use a mammalian target of rapamycin (mTOR) inhibitor only if unfit for embolisation or surgery.

ALGORITHM

1. **Spontaneous retroperitoneal haemorrhage** (Wunderlich): resuscitate, then selective arterial embolisation (**SAE**). Nephrectomy only if SAE is unavailable or unsuitable.
2. **Symptomatic** (pain, haematuria, barriers to care): SAE, then nephron-sparing surgery (NSS) or RN; ablation in selected cases.
3. **Asymptomatic sporadic**: if **>6 cm**, aneurysm **≥5 mm**, growth **>0.25 cm/yr**, or childbearing age/pregnancy, then treat; otherwise surveil.
4. **Tuberous sclerosis (TSC)**: if **>3 cm** or growing, then everolimus first line; SAE or kidney-sparing resection second.

KEY POINT

Why abandon 4 cm? It was poorly specific and caused overtreatment. "Barriers to care" counts because surveillance needs access to imaging and emergency care.

5 · OTHER BENIGN EPITHELIAL AND MESENCHYMAL TUMOURS

- **Papillary adenoma** is **≤15 mm** and cannot metastasise, so it is acceptable in a donor kidney.
- **Metanephric adenoma** (women in their 50s, polycythaemia) is **WT1 positive, CK7 negative**; excise it by NSS.
- **Mixed epithelial and stromal tumour and adult cystic nephroma** form one family and are managed by Bosniak.
- **Leiomyoma** is often really a lipid-poor AML; **Melan-A** is positive only in AML.

Tumour	Key trait
Haemangioma	Sturge–Weber, Klippel–Trénaunay; usually excised
Haemangioblastoma	Von Hippel–Lindau (VHL); mimics RCC; cured by excision
Lymphangioma	Perinephric cysts; chyluria
Reninoma (juxtaglomerular cell tumour)	Young women; refractory hypertension, hypokalaemia, high renin; excision cures
Solitary fibrous tumour	~10% aggressive: age ≥55, ≥15 cm, mitoses, necrosis

6 · LANDMARK TRIAL

Trial	Design	Result	Take-home
EXIST-2	Phase III randomised trial, everolimus vs placebo, AML in TSC or lymphangioleiomyomatosis	≥50% volume reduction in **42% vs 0%**; no bleeding needing embolisation	Approved; mTOR inhibition first line in TSC

137 Malignant Renal Tumours

1 · OVERVIEW, EPIDEMIOLOGY & AETIOLOGY

Most kidney cancers are incidental, so overtreatment is the modern risk.

- Renal cell carcinoma (RCC) makes up 2–3% of adult cancers and is the most lethal common urological cancer. Male to female ratio is **2:1**, peak age 55–75 y, and **4–6% are familial**.
- Over 60% are incidental and ~70% are organ-confined.
- Risk factors: **tobacco** (best accepted; relative risk ~1.4, dose-dependent); **obesity** (the "obesity paradox" of higher BMI but lower stage); **hypertension** (the disease, not its drugs); chronic kidney disease (CKD) and end-stage renal disease (ESRD); family history (2–4×).
- In children and young adults the commonest RCC is **Xp11/TFE3 translocation RCC**. These patients do well with surgery even when node-positive, so do a formal lymph node dissection (LND)

at radical nephrectomy (RN).

2 · CLASSIFICATION & PATHOLOGY

WHO 2022 classification (key groups)

Group	Entities
Renal cell tumours	Clear cell; multilocular cystic neoplasm of low malignant potential; papillary; chromophobe; other oncocytic; RCC not otherwise specified (NOS)
Molecularly defined (new)	TFE3/TFEB-rearranged; ELOC-mutated; fumarate hydratase (FH)-deficient; succinate dehydrogenase (SDH)-deficient; ALK-rearranged; SMARCB1-deficient medullary
Other	Collecting duct; mucinous tubular & spindle cell; tubulocystic; acquired cystic disease-associated; eosinophilic solid & cystic; clear cell papillary tumour
Non-RCC malignant	Urothelial/squamous; sarcomas; nephroblastic; neuroendocrine; lymphoma; **epithelioid angiomyolipoma**; metastases

> **KEY POINT**
> Trap: epithelioid angiomyolipoma (AML) is malignant, whereas classic AML, papillary adenoma and oncocytoma are benign. The papillary type 1/2 split was **abandoned** because "type 2" was several diseases (MET-altered, MiT translocation, FH-deficient), so WHO moved to molecular definitions.

Gross features

- RCC has a **pseudocapsule** of compressed parenchyma and is not infiltrative, except collecting duct and sarcomatoid tumours. Cystic degeneration occurs in 10–25% (better prognosis).
- Venous involvement occurs in ~10% and can reach the right atrium.
- Sporadic RCC is bilateral in 2–4% and multicentric in 10–20%; occult satellites are the main cause of ipsilateral recurrence after partial nephrectomy (PN).

Subtype (share)	Genetics	Hallmarks	Behaviour
Clear cell (70–80%)	3p loss/VHL (>90% 3p altered); +5q; PBRM1, SETD2, BAP1	Hypervascular; clear cytoplasm; carbonic anhydrase IX (CA-IX) positive; sarcomatoid 5–8%	Most venous invasion; worse when localised, better when metastatic
Papillary (10–15%)	**Trisomy 7, 17**, loss of Y; MET altered 81%	Hypovascular; CK7, AMACR; multicentric up to 40%	Good (5-y cancer-specific survival (CSS) 86–92%); common in acquired cystic disease
Chromophobe (3–5%)	Losses 1, 2, 6, 10, 13, 17; TP53, PTEN; FLCN in Birt–Hogg–Dubé (BHD)	"Plant cells", perinuclear halo, raisinoid nuclei; diffuse CK7	From intercalated cells; >90% cancer-free at 5 y unless sarcomatoid
Clear cell papillary tumour (<5%)	—	Linear papillae; CK7; **cup-like CA-IX**	Seen in ESRD and VHL; now benign
Collecting duct (Bellini)	Multiple losses	Central, infiltrative, desmoplastic; PAX8 positive; diagnosis of exclusion	Poor; urothelial-like, so may respond to cisplatin/gemcitabine
SMARCB1-deficient medullary	**Sickle cell trait**	Young Black patients; infiltrative	Dismal, median 8–13 mo
FH-deficient	FH (1q42–43)	Macronucleolus with perinucleolar halo; FH loss	Metastasises early
MiT translocation	Xp11.2 TFE3, TFEB	Clear cells with papillae; FISH confirms	40% of paediatric RCC; late nodal recurrence

Subtype (share)	Genetics	Hallmarks	Behaviour
SDH-deficient	SDH genes	Vacuolated eosinophilic cells; SDHB loss	Young; usually favourable; often germline
ALK-rearranged	ALK (2p23)	Test ALK before calling a tumour "unclassified"	Responds to ALK inhibitors in some
Unclassified (1–5%)	—	Shrinking catch-all	Poor

- **Indolent types:** multilocular cystic, tubulocystic, mucinous tubular & spindle cell, eosinophilic solid & cystic, ELOC and clear cell papillary. **Lethal types:** medullary, FH-deficient, collecting duct, unclassified and sarcomatoid.
- Because chromophobe and oncocytoma are hard to separate, WHO 2022 added "oncocytic renal neoplasm of low malignant potential".
- **Sarcomatoid** change (1–5%) is dedifferentiation of any subtype, with median survival <1 y. Sarcomatoid or rhabdoid features mean automatic grade 4 and must be reported in every RCC.

> **KEY POINT**
>
> Localised clear cell RCC does worse than papillary or chromophobe because it is intrinsically aggressive. Yet metastatic clear cell does better, because VHL-driven angiogenesis and immunogenicity make it respond to tyrosine kinase inhibitors (TKIs) and immune checkpoint inhibitors (ICIs).

WHO/ISUP grading: replaced Fuhrman; validated for clear cell & papillary

Grade	Criterion
1	Nucleoli absent/inconspicuous, basophilic at ×400
2	Nucleoli conspicuous, eosinophilic at ×400; not prominent at ×100
3	Nucleoli conspicuous, eosinophilic at **×100**
4	Extreme pleomorphism, giant cells, **rhabdoid/sarcomatoid**

Nuclear grade is an independent prognostic factor.

3 · MOLECULAR BIOLOGY & HEREDITARY SYNDROMES

The VHL pathway explains clear cell vascularity and drug targets.

P2 P1 *VHL gene and von Hippel–Lindau disease; Familial malignancies in urology — VHL–HIF pathway, two-hit, VHL lesions, genotype–phenotype, VHL screening, 3-cm rule, belzutifan; VHL/HPRC/HLRCC/BHD/TSC genes and features*

Not in the paper answer:

- BAP1 mutation (up to 14% of sporadic clear cell) gives high grade and rapid growth, so these tumours are unsuitable for active surveillance (AS).
- Loss of **CDKN2A** (9p) carries a poor prognosis in all subtypes.

ALGORITHM

1. Normoxia: the VHL protein joins elongin B/C, cullin-2 and RBX1 to form an E3 ligase, which degrades hypoxia-inducible factor (HIF)-1α/2α.
2. If VHL is lost (or in hypoxia), **HIF-2α** accumulates and pairs with ARNT. This raises VEGF, PDGF, TGF-α, GLUT1, erythropoietin and CA-IX → hypervascular clear cell RCC.

Drug targets: VEGF ligand (bevacizumab); VEGF-receptor TKIs (sunitinib, pazopanib, axitinib, cabozantinib); mTOR (temsirolimus, everolimus); HIF-2α (**belzutifan**, FDA-approved in 2021 for VHL, ~50% response); CTLA-4 (ipilimumab); PD-1 (nivolumab, pembrolizumab); PD-L1 (atezolizumab, avelumab).

Syndrome (beyond topics)	Gene (chr)	Renal tumour	Other / management
Hereditary leiomyomatosis and RCC (HLRCC)	FH (1q42–43)	Aggressive, often **solitary**	Uterine leiomyosarcoma; **prompt excision at any size, wide margins**; PN only if a wide margin is feasible; FDG-PET finds small metastases
SDH-RCC	SDHA/B/C/D, SDHAF2	Onset ≤40 y	Paraganglioma/phaeochromocytoma, GIST; excision, PN preferred
Cowden	PTEN (10q23)	Mostly papillary (lifetime 34%)	Breast, thyroid; 3-cm rule
Tuberous sclerosis	TSC1/TSC2	AMLs; clear cell 1–3%	AML: surveil <3 cm, everolimus 3–5 cm, embolise or excise ≥5 cm
BAP1	BAP1 (3p21)	High-grade clear cell	Uveal/cutaneous melanoma, mesothelioma; nephron-sparing excision
MiTF-associated	MiTF p.E318K	RCC >5×	Melanoma; management undefined

4 · CLINICAL PRESENTATION

- The **classic triad** (flank pain, gross haematuria, mass) is rare and means advanced disease. Other clues are weight loss, bone pain, a **non-reducing or right-sided varicocele**, and leg oedema from inferior vena cava (IVC) obstruction.
- Over 50% of spontaneous perirenal haematomas are due to tumour, so repeat CT later.

Paraneoplastic syndromes (10–20%)

Syndrome	Frequency / mechanism	Key point
Raised erythrocyte sedimentation rate (ESR)	Commonest	Marker of advanced disease
Hypercalcaemia	Up to 13%; parathyroid hormone-related protein (PTHrP) or osteolytic metastases	**The only one managed medically**: hydration, then furosemide; zoledronic acid 4 mg IV 4-weekly (withhold in renal insufficiency); denosumab if refractory; steroids, calcitonin. Definitive: nephrectomy with or without metastasectomy
Hypertension	Renin, artery compression, arteriovenous fistula	—
Polycythaemia	Erythropoietin	—
Stauffer syndrome	3–20%; interleukin-6	Non-metastatic liver dysfunction (ALP raised almost always); exclude liver metastases; normalises after nephrectomy in 60–70%; persistence means viable tumour and poor prognosis
Others	Rare	Cushing, hyperglycaemia, galactorrhoea, neuromyopathy, clotting disorders

- Paraneoplastic syndromes resolve only with excision or systemic therapy, so persistence after nephrectomy means residual disease.

5 · DIAGNOSIS

Imaging sorts a mass into cyst, fat or enhancing tumour.

P4 *Radiology for solid renal masses — CT phases and HU thresholds, MRI, CEUS, sestamibi, girentuximab, FDG-PET, biopsy yield, venous thrombus CT sensitivity, R.E.N.A.L./PADUA*

Not in the paper answer:

- The strongest predictors of malignancy are **male sex (~3×)** and **size**: 20–30% of masses ≤4 cm and ~40% of ≤2 cm are benign, while few ≥7 cm are benign.
- A cT1 mass is ~20% benign, ~60% indolent and ~20% potentially aggressive.
- Renal mass biopsy (RMB): **multiple cores over fine-needle aspiration**; haematoma ~5%, specificity 94%. A non-malignant result does not prove the mass is benign.
- **Multiphase CT** is the single most important test: enhancement **>15–20 Hounsfield units (HU)** suggests RCC, and fat **<−20 HU** means AML. MRI is best for IVC thrombus and organ invasion.

ALGORITHM

1. If simple cyst, then observe.
2. If complex or solid: eGFR >45 → CT or MRI with/without contrast; eGFR 30–45 → hydrate, then the same; eGFR <30 → non-contrast CT or gadolinium MRI; contrast allergy → ultrasound (USG), contrast USG or MRI.
3. Then: fat → AML; complex cyst → Bosniak; solid enhancing → management options, with or without RMB.

- **Bosniak** (see Ch 136): malignancy is 3–10% in IIF, ~50% in III and 75–90% in IV. The key discriminator is unequivocal **measurable enhancement**. Avoid ablation and biopsy of cystic masses; AS is acceptable for III/IV.
- Do RMB when the result may change management, or when lymphoma, metastasis or infection is suspected. Do not biopsy patients who will be managed the same way regardless. Biopsy before ablation.
- **Screening:** there is none for the population or for autosomal dominant polycystic kidney disease. In ESRD, screen with USG or CT from the 3rd year of dialysis if fit, and continue after transplant.
- Offer genetic counselling for RCC at **≤46 y**, multifocal or bilateral tumours, or a suggestive history or pathology.

6 · STAGING & PROGNOSIS

Pathological stage is the strongest predictor of outcome.

TNM (AJCC 8th)	Definition	5-y survival
T1a / T1b	≤4 / >4–7 cm, confined	95–100 / 85–95%
T2a / T2b	>7–10 / >10 cm	65–85 / 55–75%
T3a	Renal vein/segmental branches, **pelvicalyceal system**, perirenal/sinus fat, within Gerota's	45–75%
T3b	IVC below diaphragm	40–55%
T3c	IVC above diaphragm or **caval wall invasion**	20–45%
T4	Beyond Gerota's, incl. **contiguous ipsilateral adrenal**	0–35%
N1 / M1	Regional node / distant (non-contiguous adrenal = M1)	0–30%

Stage groups: I is T1N0; II is T2N0; III is T1–2N1 or T3 any N; IV is T4 or M1. The 8th edition moved pelvicalyceal invasion to T3a.

KEY POINT

TNM limitation: stage is set by the single worst feature, so co-existing adverse features (e.g. adrenal invasion plus a supradiaphragmatic thrombus) are lost. Record all of them and use integrated nomograms.

- **Work-up:** abdominal CT and chest X-ray (chest CT if symptoms or high risk). Bone scan only with raised ALP, bone pain or poor performance status (PS); no PET. Treat an abnormal adrenal as

involved.

- **Pathological stage** is the most important factor, then grade, size, subtype, necrosis and sarcomatoid change. Node-positive 5-y survival is 5–30%, and with a thrombus but otherwise confined disease it is 45–69%.
- Venous wall invasion is as important as thrombus level, and collecting-system invasion is adverse.
- Nomograms: **UISS** uses TNM, grade and ECOG PS; **SSIGN** (Mayo) uses Stage, Size, Grade and Necrosis; **Leibovich** predicts recurrence in clear cell RCC.

7 · MANAGEMENT OF LOCALISED RCC

Every option trades cancer control against kidney function.

P1 **P2** *Embryology and anatomy of kidney; nephron-sparing surgery in small renal mass and renal malignancy; Nephron-sparing surgery / partial nephrectomy in RCC — imperative/relative/elective PN, AUA RN rule (GFR >45), surgical vs medical CKD, PN steps, enucleation, solitary kidney, AS/TA criteria, 3-cm rule, EORTC 30904, DISSRM*

Not in the paper answer:

- AUA 2021: refer to nephrology if GFR <45, confirmed proteinuria, diabetic CKD, or expected post-op GFR <30. Do LND for clinically positive nodes and adrenalectomy only if suspicious.
- CKD follows RN in 65% vs PN in 20%; surgical CKD progresses ~0.7%/y vs medical CKD ~4.7%/y. **Predicted GFR after RN ≈ global GFR × contralateral split function × 1.25**.
- **Triggers to intervene on AS**: >3 cm, growth >5 mm/y, stage progression, adverse biopsy, symptoms.
- On AS, small masses grow 0.1–0.3 cm/y and metastasise in <2%; image every 3–6 mo.
- PN function depends first on parenchymal quality, then on the **amount of vascularised parenchyma preserved (the main modifiable factor)**, and only secondarily on ischaemia.
- Leak and bleeding are commoner after PN than RN (steps: Ch 142).
- In a solitary kidney at least 20–30% of the kidney is needed. If bilateral tumours make PN impossible, options are RN with dialysis or neoadjuvant TKI.
- Enucleation is controversial in sporadic RCC because 40–50% infiltrate the pseudocapsule.

Radical nephrectomy

- Robson's principles (early vascular ligation, removal outside Gerota's, adrenalectomy, extended LND) are now modified: perifascial dissection is still essential (~25% of cT1b/T2 invade fat), but there is **no routine adrenalectomy** (unless enlarged on imaging or extensive tumour) and **no routine extended LND**.
- Laparoscopic RN is standard without gross invasion, major venous involvement or bulky nodes; open surgery uses a subcostal or thoracoabdominal approach (see Ch 141).

Thermal ablation (see Ch 143)

- Thermal ablation (TA) works best <2.5–3 cm and gives no pathological staging. Cryoablation is lethal at ≤−20 °C with a double freeze–thaw; neither cryoablation nor radiofrequency is superior.
- Recurrence is mostly salvaged by repeat TA, and nodular enhancement means recurrence.

Familial RCC: 3-cm rule (National Cancer Institute)

- VHL tumours <3 cm had no metastases at ~5 y, while ~25% metastasise when larger. **HLRCC is the exception**.
- Recurrence is near-universal, so surveillance is lifelong. **Exclude phaeochromocytoma** and central nervous system haemangioblastoma before surgery.

8 · LOCALLY ADVANCED RCC

Even with a caval thrombus, complete surgery can cure.

IVC tumour thrombus (venous involvement 4–10%; cure 45–70%)

Level	Extent	T	Vascular control
Renal vein	Vein/branches	T3a	Standard RN
I	At renal vein ostium	T3b	Satinsky clamp/stapler; keep ≥50% lumen; patch if needed
II	Below hepatic veins	T3b	Clamp infrarenal IVC, then contralateral renal vein, then suprarenal IVC; occlude lumbars
III	Intrahepatic, below diaphragm	T3b	Mobilise liver and milk thrombus below hepatic veins; or suprahepatic clamp with Pringle; venovenous bypass often
IV	Above diaphragm	T3c	Cardiopulmonary bypass with or without hypothermic arrest; avoid opening the heart if thrombus can be brought below the atrium
Any, wall invasion	Caval wall	T3c	Resect to negative margins; graft/reconstruct

- Image close to surgery, because the thrombus grows. **No routine renal artery embolisation** (more deaths). Do coronary angiography before bypass (levels III–IV) and use intraoperative transoesophageal echocardiography.
- Level and wall invasion predict survival, and level IV without metastasis is curable. Mortality is 5–10% for supradiaphragmatic thrombus.

Locally invasive (T4)

- Only ~40% of suspected invasion is confirmed. En bloc resection is the only cure, but prognosis is poor.

Lymph node dissection

- Only 30–43% of enlarged nodes are malignant. With ≥2 risk factors (≥10 cm, cT3/T4, grade 3–4, sarcomatoid change, necrosis), >40% are node-positive.
- LND gave no survival benefit (EORTC 30881), so it is done for suspicious nodes, mainly for staging.

9 · RECURRENCE, ADJUVANT THERAPY & FOLLOW-UP

A third of recurrences come late; one adjuvant drug works.

Local recurrence	After RN	After PN	After TA
Rate	2–4%	1.4–10%	Cryo 3–10%, radiofrequency ablation 5–20%
Site	Fossa, adrenal, vein stump, nodes	Remnant, usually away from bed	Ablation zone
Management	Only 20–40% isolated, so do a full work-up; en bloc resection gives 30–50% long-term cancer-free	Repeat PN, completion nephrectomy, TA, AS	Repeat TA, AS, salvage surgery

- A positive PN margin is usually surveilled; completion nephrectomy is only for gross involvement.
- About 20–35% metastasise after "curative" surgery, but cytokines, vaccines and most TKIs failed as adjuvants. **Adjuvant pembrolizumab** is FDA-approved for high risk.

Follow-up (AUA risk groups)

Risk	Definition	Chest imaging
Low	pT1 grade 1–2	X-ray
Intermediate	pT1 grade 3–4 or pT2	X-ray
High	pT3	CT
Very high	pT4, pN1, sarcomatoid/rhabdoid, gross positive margin	CT

- Imaging starts at 3–6 mo and intensifies with risk; very high risk is followed for up to 10 y. A microscopic positive margin moves the patient up one level. **30% of recurrences occur after 5 y**.
- Benign pathology needs no imaging. After TA, do contrast imaging within 6 mo, then follow the intermediate schedule.

10 · OTHER RENAL MALIGNANCIES

Lymphoma and metastases need biopsy and drugs, not nephrectomy.

Tumour	Clues	Management
Sarcoma (1–2%)	**Leiomyosarcoma commonest** (50–60%); capsule/sinus origin, huge, no nodes, hypovascular	Wide en bloc excision, negative margins; the first resection is the best chance; margin and grade give prognosis
Lymphoma	Non-Hodgkin; multiple masses (45%); massive adenopathy, splenomegaly; prior lymphoma means recurrence until proven otherwise	**Biopsy, avoid nephrectomy**; R-CHOP with or without radiotherapy
Metastases	Commonest renal malignancy (12% at autopsy); lung primary commonest; multifocal, 5–30 HU	Biopsy; systemic therapy; nephrectomy only for refractory bleeding
Carcinoid	**Horseshoe kidney (82×)**	Excision; good prognosis
Small cell	Usually advanced	Surgery and platinum
Primitive neuroectodermal tumour	Ewing family; CD99; t(11;22)	Multimodal; poor
Adult Wilms	3% of Wilms; triphasic	Paediatric protocols; worse than in children

- In a suspected sarcoma, any RCC feature excludes it (the differential is sarcomatoid RCC).

> **KEY POINT**
>
> Non-RCC malignancies are almost all **hypovascular**. The key decision is surgical (sarcoma, carcinoid) vs medical (lymphoma, metastasis: biopsy, don't resect). RCC nodes are interaortocaval (right) or para-aortic (left), so nodes elsewhere suggest lymphoma.

11 · LANDMARK TRIALS

Trial	Design	Result	Take-home
EORTC 30904	Randomised trial (RCT), RN vs elective PN, ≤5 cm, normal contralateral kidney	PN better function; 10-y overall survival (OS) better with RN (81 vs 76%)	Flawed; PN's functional gain may be smaller than believed
EORTC 30881	RCT, RN with or without LND	No survival benefit	Underpowered, low-risk (4% node-positive); LND for staging
S-TRAC	Sunitinib vs placebo, high-risk clear cell	Disease-free survival (DFS) 6.8 vs 5.6 y; no OS gain	Only positive adjuvant TKI trial
KEYNOTE-564	Pembrolizumab vs placebo, high-risk	DFS hazard ratio 0.63	FDA 2021; only positive adjuvant ICI trial
DISSRM	Registry, AS vs intervention, small masses	1 death/metastasis on AS; 24% delayed intervention	AS safe for small masses

138 Urothelial Tumours of the Upper Urinary Tract and Ureter

1 · EPIDEMIOLOGY & RISK FACTORS

UTUC is often invasive at diagnosis, and its causes act as a field effect.

P3 P2 *Upper Tract Urothelial Carcinoma: Diagnosis, Staging, Minimally Invasive, Endoscopic and Radical Management; Etiopathogenesis and management of Upper tract TCC — epidemiology, tobacco/aristolochic acid/Lynch/occupation/phenacetin/arsenic, seeding vs field effect, CTU, cytology, URS biopsy, TNM, EAU risk groups, management*

Not in the paper answer:

- Upper tract urothelial carcinoma (UTUC) is **5–10%** of urothelial cancers, but **40%** of renal neoplasms in the Balkans. Concurrent bladder cancer occurs in **17%**, ~60% are invasive at diagnosis, and multifocality is **10–20%**.
- After UTUC, the bladder recurs in **22–47%**.
- **EAU: screen for Lynch syndrome (LS) at the first interview** if age <60, a personal LS-spectrum cancer, a first-degree relative (FDR) <50 with LS cancer, or two FDRs with LS cancer.

> **KEY POINT**
>
> **Why UTUC presents invasive:** the muscle layer is thin, so tumours reach muscle, parenchyma or periureteric fat earlier than bladder tumours.

- **Smoking** carries a relative risk (RR) of **2.5–7**. Chronic infection or stones (RR 1.5–2) favour squamous cell carcinoma (SCC). Cyclophosphamide, long-term anthranoid laxatives and pelvic radiotherapy are iatrogenic causes.

> **KEY POINT**
>
> Aristolochic acid, arsenic, phenacetin and Lynch all behave as a **field disease** (multiple, bilateral recurrences). This favours kidney-sparing treatment, but Balkan endemic nephropathy patients often have renal insufficiency.

- **Lynch**: UTUC risk is ≈**14×**, and ≈**75×** in MSH2 carriers. If mismatch repair proteins are lost on immunohistochemistry, do germline testing.

2 · PATHOLOGY & MOLECULAR BIOLOGY

- Over 90% are urothelial carcinoma. Variant histology (≈**25%**: squamous, glandular, sarcomatoid, micropapillary, neuroendocrine) is aggressive, and **micropapillary** is independently adverse.
- Ureteric inverted papilloma harbours malignancy in ≈**18%**, so follow it for ≥2 yr.
- SCC (0.7–7%) is commoner in the pelvis and invasive at presentation; adenocarcinoma is rare and linked to stones.
- Microsatellite instability distinguishes UTUC from bladder cancer, is prognostic and flags LS; the **RAS-mutated** subtype does worst.

3 · DIAGNOSIS

Biopsy gives grade but not stage.

- Haematuria is the commonest presentation, then flank pain. **Always do cystoscopy**; positive cytology with normal cystoscopy (including urethra) means the upper tracts must be evaluated.
- **CT urography (CTU)** is first line (EAU prefers it to MR urography): sensitivity ≈**92%**, specificity ≈95%, but flat lesions are missed. Hydronephrosis means an invasive ureteric tumour in up to 80%.
- For MR urography, avoid gadolinium if creatinine clearance <30 mL/min. Do a split-function scan before radical surgery.

- Cytology detects ≈65% overall (G1 20%, G3 75%); selective washing beats voided urine, and it should be **collected before contrast**.
- Flexible ureteroscopy (URS) with biopsy raises accuracy to ≈90%; biopsy grade is ≈90% accurate, but visual grading only ≈70%.

> **KEY POINT**
>
> **Why URS biopsy cannot stage:** samples are small and shallow, so a Ta biopsy is often T3 at resection. Stage is **inferred** from grade, appearance, hydronephrosis and cytology, which is why kidney-sparing patients need intensive follow-up.

- **Diagnostic URS indications:** uncertainty, planned kidney-sparing treatment, solitary kidney. **Pitfall:** URS before radical nephroureterectomy (RNU) increases intravesical recurrence without changing survival, so avoid it when CTU and cytology already justify RNU.

4 · STAGING & GRADING

Pelvic T3 vs T4 is a classic trap.

TNM 8th	Definition
Ta / Tis	Non-invasive papillary / CIS
T1	Subepithelial connective tissue
T2	Muscularis
T3	Pelvis: peripelvic fat or **renal parenchyma**; ureter: periureteric fat. pT3a = microscopic parenchymal; pT3b = macroscopic or peripelvic fat (worse)
T4	Adjacent organs, or through kidney into perinephric fat
N1 / N2	Single node ≤2 cm / single >2 cm or multiple

- AJCC stages: I = T1; II = T2; III = T3; **IV = T4, any N+ or M1**.
- Pelvic and upper ureteric tumours drain to hilar and retroperitoneal nodes, and lower ureteric tumours to pelvic nodes. WHO grades are PUNLMP (managed as low grade), low and high grade.

5 · PROGNOSIS & RISK STRATIFICATION

The risk group decides between kidney-sparing treatment and RNU.

- **Stage matters most**, then grade. SEER 5-yr cancer-specific survival after RNU: ≤T1 86%, pT2 77%, pT3 63%, pT4/N+ 39%.
- **Ureteric and multifocal tumours do worse** than pelvic ones at the same stage, because there is no parenchymal barrier and the periureteric lymphovascular plexus is rich.
- Age alone is never a reason to deny RNU. Do **RNU within 12 wk** for invasive disease. Size **2 cm** is the best cut-off for ≥pT2.
- Post-op adverse factors include nodal disease, lymphovascular invasion (never seen on biopsy, so report it on every specimen), positive margins, necrosis, sessile growth, CIS and variant histology.

Low risk — ALL	High risk — ANY ONE (EAU 2020)
Unifocal	Multifocal
<2 cm	>2 cm
Low-grade cytology	High-grade cytology
Low-grade URS biopsy	High-grade URS biopsy
No invasive aspect on CTU	Hydronephrosis
—	Previous radical cystectomy; variant histology

KEY POINT

Every pre-op factor is a stand-in for invasion, so one adverse feature means high risk. **Trap:** a small, low-grade, unifocal tumour **with hydronephrosis is high risk**.

6 · RECURRENCE PATTERNS

- Bladder recurrence after UTUC is 15–75% at 5 yr, usually within 2 yr.
- Upper tract tumours after bladder cancer occur in 2–4%, late (median >3 yr). About **30%** of bacillus Calmette–Guérin (BCG)-treated bladder CIS (or cystectomy for BCG-refractory disease) develop UTUC.

KEY POINT

Asymmetric recurrence: antegrade seeding makes bladder recurrences common, early and monoclonal, while the field effect gives multifocal tumours of different clones. Both operate.

- Intravesical recurrence after RNU is linked to male sex, prior bladder cancer, ureteric or multifocal tumours, laparoscopic or extravesical cuff technique, positive margins and diagnostic URS.
- **Pan-urothelial disease** (bladder plus ≥2 extravesical sites) has no evidence-based algorithm; note that cystectomy before RNU may lose cisplatin eligibility.

7 · MANAGEMENT OVERVIEW & FOLLOW-UP

ALGORITHM

1. Work-up: cystoscopy and CTU, with or without URS biopsy and cytology; assess contralateral function; screen for LS; then assign the risk group.
2. If **low risk**, then kidney-sparing treatment (endoscopic or segmental) with intensive surveillance.
3. If **high risk**, then RNU with bladder cuff, with or without LND, within 12 wk.
4. If **cN+**, then systemic chemotherapy first, and surgery only if there is a response.
5. After RNU, give a single intravesical chemotherapy dose; all patients have cystoscopic follow-up.

- Pre-op nomograms (≥pT2 or non-organ-confined) select patients for neoadjuvant chemotherapy, given before RNU because nephrectomy often removes cisplatin eligibility.
- Lymph node dissection (LND) gives the best staging, but its curative role is debated (technique and POUT: Ch 139).
- Follow-up: cystoscopy for all. After kidney-sparing treatment, add URS, in-situ cytology and CTU. After high-risk bladder cancer (BCG-treated CIS, BCG-refractory), do yearly upper-tract CT. Surveillance is prolonged because the risk is constant.

139 Surgical Management of Upper Urinary Tract Urothelial Tumours

Epidemiology, risk factors, pathology, staging and imaging: Ch 138

1 · PRINCIPLES & PATIENT SELECTION

Tumour risk and renal reserve together decide between removing and saving the kidney.

P3 *Upper Tract Urothelial Carcinoma (including TCC of Ureter): Diagnosis, Staging, Minimally Invasive, Endoscopic and Radical Management — AUA/SUO favourable/unfavourable groups and ablation role, elective vs imperative KSM, URS and percutaneous steps, lasers, segmental options, RNU principles, abandoned cuff techniques, topical BCG/UGN-101, follow-up*

Not in the paper answer:

- Endoscopy carries ~3× more reinterventions than RNU; ipsilateral recurrence after segmental resection is 33–55%; no routine adrenalectomy.
- **Standard** treatment of upper tract urothelial carcinoma (UTUC) is radical nephroureterectomy (RNU) with bladder cuff, if the other kidney is normal and the patient fit.
- UTUC is **unforgiving of spillage**, so every approach keeps the collecting system closed.
- Work-up is CT or MR urogram, cystoscopy, then ureteroscopy (URS) with biopsy and selective cytology; URS raises accuracy from ≈75% to 85–90%.
- Keep diagnostic endoscopy brief and low-pressure to avoid seeding.
- Low-grade (LG) tumours are mostly Ta/T1; high-grade (HG) tumours are mostly invasive.

*AUA/SUO risk group	Features	Ablation	Systemic therapy
Low – favourable	LG, negative cytology, unifocal, papillary, no obstruction/invasion	**Preferred**	No
Low – unfavourable	LG with obstruction, multifocality or prior lower-tract tumour	May be offered	No
High – favourable	HG, unifocal, papillary, no invasion/obstruction	Rare, select cases	Neoadjuvant/adjuvant
High – unfavourable	HG cytology, invasion on imaging, obstruction, suspicious nodes, multifocal, sessile/flat	Palliation only	Neoadjuvant/adjuvant

*American Urological Association / Society of Urologic Oncology.

KEY POINT

Low grade does not mean automatic ablation: obstruction or multifocality makes an LG tumour unfavourable, where RNU keeps a role.

KEY POINT

Two axes: tumour risk and renal reserve. High risk with adequate reserve means extirpation; low risk means preservation. Imperative indications are where they conflict.

ALGORITHM

1. Imaging suspicion: CT or MR urogram.
2. Endoscopy with biopsy and cytology; assign risk group; shared decision.
3. If high risk with adequate glomerular filtration rate (GFR), then RNU with or without neoadjuvant chemotherapy.
4. If low risk or poor GFR, then kidney-sparing management (KSM).

- Imperative KSM must not significantly compromise cancer control; dialysis risk is weighed against oncological risk.
- Unfavourable lower ureteric tumours need distal ureterectomy, not endoscopy.
- **Expectant management** (AUA/SUO option) is only for major comorbidity or high risk of end-stage renal disease (ESRD); untreated UTUC progresses.

2 · ENDOSCOPIC (KIDNEY-SPARING) MANAGEMENT

- **Grade** is the key selector and the strongest predictor of recurrence. AUA/SUO: tumours **<1.5 cm**; caution at ≥1.5 cm.
- For a bulky or suspected HG tumour, use a **staged approach**: biopsy and partial ablation only, avoiding perforation, and complete it once pathology confirms suitability.
- Versus RNU, survival is similar but there are **~3× more reinterventions** and lifelong surveillance.

	Retrograde ureteroscopic	Antegrade percutaneous
System	**Closed**	**Open** – tract seeding ≈0.3%, all HG

	Retrograde ureteroscopic	Antegrade percutaneous
Best for	Low-volume ureteric, small renal	Large pelvic/upper ureteric, lower-pole calyx, diversion
Staging	Not possible	Possible – base sent separately
Morbidity	Low	Higher (≈25%), admission
Recurrence	Higher bladder and upper-tract	Lower
Second look / topical therapy	Repeat URS; catheter or stent reflux	Via retained nephrostomy

- **URS:** saline barbotage improves cytology, and a tight ureter is pre-stented rather than forcefully dilated. The usual method is a **3-Fr biopsy, then laser**; resection never goes beyond the lamina propria.
- A tumour protruding from the orifice needs ablation or resection of the whole intramural ureter.
- **Never fulgurate circumferentially**, because a stricture follows.
- Lasers: holmium:YAG <0.5 mm, ideal for the ureter; neodymium:YAG 5–6 mm, with transmural risk; thulium <0.2 mm, with strong coagulation.
- **Percutaneous:** puncture the calyx distal to a peripheral tumour, never through it, and dilate to 30 Fr. Resect with cold-cup forceps or a **bipolar** resectoscope, since monopolar risks hypo-osmotic absorption.
- A **nephrostomy is always left** for second look and topical therapy. HG or invasive histology means RNU.
- In a diversion, use flexible or antegrade endoscopy.

> **KEY POINT**
>
> Does URS spread tumour? Prior ureteroscopy increases bladder recurrence (seeding), but pre-RNU URS has not increased metastasis. A single post-op intravesical instillation mitigates this.

3 · SEGMENTAL URETERECTOMY

- It saves nephrons **and** gives a staging specimen; in selected patients it equals RNU.
- **URS of the entire tract above is mandatory**. For HG, add ipsilateral pelvic lymph node dissection (LND).
- Ipsilateral recurrence is **33–55%**, mostly distal and lifelong.
- Segmental resection with uretero-ureterostomy is rare and needs 1–2 cm margins with frozen section.
- Distal ureterectomy with reimplant suits LG low-stage tumours, or selected HG distal ones when the kidney must be kept.
- **Boari flap:** contraindicated by a small or irradiated bladder. The base is ≥2 cm wider than the apex, and the width ≥3× the ureteric diameter.
- **Ileal ureter:** 20–25 cm of ileum ≥15 cm from the ileocaecal valve, isoperistaltic.

Manoeuvre	Length gained	Condition
Psoas hitch	≈5 cm	Mobile bladder; avoid nerve entrapment
Boari flap with hitch	Further 10–15 cm	Adequate capacity
Renal descensus with nephropexy	Up to 8–10 cm	Better on left (longer renal vein)
Ileal ureter / autotransplant	Whole ureter	Autotransplant in skilled hands

4 · POSITIVE CYTOLOGY & CARCINOMA IN SITU (CIS)

Trace a positive cytology before removing any kidney.

ALGORITHM

1. Positive voided cytology: repeat it.
2. CT urogram and bladder biopsies. If the bladder is positive, then treat the bladder.
3. If still positive with a negative bladder (or after BCG "6+1"), then selective upper-tract cytology, retrograde pyelography (RPG), prostatic urethral biopsy (men) and URS.

- Upper-tract CIS is a diagnosis of exclusion: persistently positive selective cytology with normal URS and imaging.
- **No RNU for cytology alone**, because of false positives and later contralateral disease.
- AUA/SUO: **6-week BCG induction**, preferably via nephrostomy; RNU if more than surface disease is proven.
- CIS with HG papillary UTUC needs radical or multimodal treatment.

5 · TOPICAL THERAPY IN THE UPPER TRACT

- Indications are solitary kidney, bilateral UTUC and ESRD risk. Routes are nephrostomy (≤25 cm H_2O), ureteric catheter or stent reflux.
- Always use **low pressure and sterile urine**, as sepsis deaths are reported.
- Adjuvant instillation after papillary ablation has **no proven benefit**. BCG is acceptable after complete ablation of HG favourable Ta/T1; its commonest complication is renal granulomas.
- **UGN-101 (Jelmyto)** is mitomycin C (MMC) in a reverse-thermal gel, FDA-approved in 2020 for LG UTUC **not endoscopically manageable**. Stricture occurs in ≈44%.

6 · RADICAL NEPHROURETERECTOMY

Remove the whole ureter and cuff without spillage.

- **Indications:** non-metastatic high-risk pelvic or proximal ureteric tumour (HG, obstruction, parenchymal invasion), with cuff and LND; or LG with unfavourable features or failed or infeasible endoscopy.
- **Kidney first, cuff last. Clip the ureter early**, control the hilum before manipulation and stay outside Gerota's fascia. **Adrenalectomy is not routine.**
- Remove the **entire intramural ureter and orifice**: a retained stump recurs in **30–75%** and cannot be surveyed. After bowel diversion, excise the ureteroenteric anastomosis.
- Extravesical or transvesical excision is acceptable; lasso, pluck, stripping and stapling are abandoned because of spillage and positive margins. Close in two layers.
- Laparoscopic or robotic: wide perihilar margin and extract **intact in a bag**.
- Ureteric blood supply is anteromedial in the upper third, medial in the middle and **lateral** in the lower third.

	Open	Laparoscopic	Robotic
Oncology	Reference	Equivalent	Comparable
Morbidity	Highest	Less blood loss, shorter stay	Lowest; fewer positive margins
Caveat	—	Possibly more bladder recurrence; less favoured for T3/T4	Longest; more LND done

KEY POINT

Principle beats approach: port-site seeding and carcinomatosis follow spillage, not the approach.

Lymphadenectomy

P4 *Role of neoadjuvant therapy and Lymphadenectomy for muscle invasive bladder cancer and upper tract urothelial cell cancer? — UTUC templates (right interaortocaval), ≥8 nodes, chylous leak; neoadjuvant rationale, EA8141, AUA/SUO CrCl <50, POUT*

Not in the paper answer:

- Outcome ranks pN0 better than pNx, better than pN+; the prognostic cliff is T2 to T3.
- LND has value in **T2–T4**, especially for pelvis and proximal ureter, and the surgeon, not the approach, sets the yield.
- Caudal limits: renal pelvis to the inferior mesenteric artery; upper two-thirds of ureter to the aortic bifurcation; lower third to the pelvis.

7 · PERIOPERATIVE ADJUNCTS

- Bladder tumours follow in 30–50% after RNU or endoscopy, mostly within 2 yr. **AUA/SUO strongly recommend** a single dose after RNU (optional after endoscopy).
- Post-op (day 3–7), do a cystogram, then give MMC or gemcitabine with a 1–2 h dwell. An **intraoperative** instillation, clamped until the ureter is clipped, needs no cystogram.
- **Renal function is the hinge**, so give platinum **before** nephrectomy. Neoadjuvant is favoured if post-op creatinine clearance (CrCl) is expected **<u><50</u>** mL/min.
- In EA8141, patients with CrCl >50 received accelerated MVAC ×4.
- Adjuvant (POUT): gemcitabine–cisplatin (CrCl >50) or gemcitabine–carboplatin (30–49). cN+ disease receives chemotherapy, then consolidative surgery.

8 · COMPLICATIONS

Procedure	Complication	Management / prevention
URS	Perforation; stricture	Stent; pre-stent tight ureter; no circumferential fulguration
Percutaneous	Bleeding, irrigant absorption, perforation, secondary ureteropelvic junction obstruction	Bipolar resection; avoid major vessels; nephrostomy
Topical therapy	Sepsis, absorption, stricture	Sterile urine, low pressure
Cuff closure	Urine leak	Two-layer closure; catheter; cystogram
RNU	Renal insufficiency	Nephrology referral

9 · FOLLOW-UP

- Every visit includes **cystoscopy**, cytology and a metabolic panel; after KSM add ipsilateral URS.
- Use CT or MR urography (RPG if no contrast) and chest imaging.

Group	Cystoscopy	Upper-tract endoscopy	Imaging
KSM low risk	3 mo, then 6–9-monthly ×2 yr, then yearly	3 mo, 6 mo, 1 yr	6–9-monthly ×2 yr, yearly to 5 yr
KSM high risk	3 mo, then 3–6-monthly ×3 yr, then yearly	3 mo, 6 mo, 1 yr	3–6-monthly ×3 yr, yearly to 5 yr
RNU <pT2N0	With cytology at 3 mo, then by grade	—	CT/MRI by 6 mo, yearly ≥5 yr
RNU ≥pT2	3 mo, 3–6-monthly ×3 yr, yearly	—	Abdomen/pelvis 3–6-monthly ×2 yr; chest CT 6–12-monthly ×5 yr

10 · LANDMARK TRIALS

Trial	Design	Result	Take-home
ODMIT-C	Randomised controlled trial (RCT), MMC 40 mg single dose at catheter removal after RNU	Bladder recurrence 28% vs 16% at 1 yr	Single-dose instillation standard

Trial	Design	Result	Take-home
Pirarubicin trial (Ito)	RCT, single pirarubicin dose within 5 days of RNU	Bladder recurrence 42% vs 17% at 2 yr	Confirms single post-RNU instillation
Laparoscopic vs open RNU	Only RCT; small	Equal 5-yr survival; less blood loss	Minimally invasive acceptable
UGN-101 phase III	Single-arm, LG UTUC	Complete response 59%; 56% disease-free at 12 mo	Gel chemoablation option
POUT	Phase III adjuvant platinum vs surveillance, pT2–4/N+	Stopped early; disease-free survival (DFS) hazard ratio **0.45**; 3-yr DFS 71% vs 46%	Adjuvant platinum standard after RNU for ≥pT2/N+

141 Contemporary Open Surgery of the Kidney

1 · ROLE & PREOPERATIVE WORK-UP

- Open surgery offers **exposure and early great-vessel control**. It suits renal cell carcinoma (RCC) with inferior vena cava (IVC) thrombus, cytoreduction, complex tumours and a hostile abdomen (minimally invasive surgery: Ch 142).
- **Renal function** decides partial (PN) versus radical nephrectomy (RN). Estimate glomerular filtration rate (GFR) with the **CKD-EPI 2021 race-free** equation, which beats MDRD.
- If GFR is **≤60** or there is significant proteinuria, ask for a perioperative nephrology consult.
- In advanced RCC, check blood count and metabolic panel (MSKCC and IMDC risk models), liver tests (**Stauffer syndrome**) and cardiopulmonary fitness.
- Contrast is generally safe if GFR is **>30**. If thrombus is suspected, MR or CT venography must be **≤1 month old**, because thrombus propagates rapidly.
- Give cefazolin ≤1 h pre-incision, pneumatic compression and subcutaneous heparin 5000 IU; no routine bowel preparation.
- **Renal artery embolisation** lets the vein be tied first when the hilum is hidden, but it is no longer routine because of postinfarction syndrome and harder surgery if delayed >48 h.

2 · SURGICAL ANATOMY & APPROACHES

Most disasters come from misread anatomy.

- At risk on the **right** are liver, duodenum and colon; on the **left**, spleen, pancreatic tail, colon and superior mesenteric artery (SMA).
- The **right renal artery** lies behind the renal vein and IVC and can be reached in the **interaortocaval space** by retracting the left renal vein (LRV) upward.
- The **SMA** leaves the front of the aorta **1–2 cm above LRV** and veers left, so it can be mistaken for the left renal artery.
- The right gonadal and adrenal veins drain into the IVC; the left ones drain into the LRV.
- Anterior exposure reflects the colon along the **white line of Toldt**. On the right, add a **Kocher** manoeuvre without cautery, because duodenal injury is devastating.
- **Cattell-Braasch** (right colon and midgut mobilisation) and partial **Mattox** (left medial visceral rotation) expose the great vessels.

Incision	Best for	Limitations
Midline	Trauma; advanced tumour or IVC thrombus (early aortic access); bilateral	Bowel mobilisation needed
Anterior subcostal	Unilateral transperitoneal	Muscle division

Incision	Best for	Limitations
Chevron	Renovascular; RN with thrombectomy; both pedicles	Extensive muscle division
Subcostal flank (below 12th rib)	Lower pole, pelviureteric junction, proximal ureter	Poor hilar access; not for PN
Supracostal (11th/12th)	RN or PN; 8-cm mini-flank above the 11th	Pleura
Transcostal (11th, rib resected)	Retroperitoneum at an old rib tip	Rib division
Thoracoabdominal, dorsal lumbotomy	Largely historical; thoracoabdominal is still an option for level III–IV thrombus	

- The **flank** position lowers venous return and lung function, so it is poor in cardiopulmonary disease; pad against rhabdomyolysis.
- In the **supracostal** approach, push the pleura up and divide the intercostals close to the **upper border of the rib** to spare the bundle. Spare the **subcostal nerve**.

3 · SIMPLE NEPHRECTOMY

- Stay **within Gerota's fascia**, on the capsule, for a benign nonfunctioning kidney (obstruction, infection, stones, reflux), and spare the adrenal.
- Benign PN suits focal atrophy, a duplex pole or a symptomatic angiomyolipoma.

4 · RADICAL NEPHRECTOMY

- Robson's operation included adrenal and extended nodes; now the **normal adrenal is spared** and lymph node dissection (LND) is selective.
- **Indications:** tumours not amenable to PN, such as a nonfunctioning kidney, tumour replacing the kidney, nodes, cytoreduction or venous thrombus.
- Staging is abdominal CT or MRI and **chest CT**; a bone scan only for symptoms or raised alkaline phosphatase.
- Biopsy only if it changes the plan, such as an infiltrative mass or before systemic therapy.
- T4 invasion means altered enhancement of the organ, not just contact.

ALGORITHM

1. **Right:** survey the abdomen; open Toldt's line down to the iliac vessels; take down the hepatic flexure; Kocher; expose the IVC.
2. Loop and palpate the renal vein; find the artery behind it; tie **artery, then vein**; remove the kidney outside Gerota's.
3. **Difficult hilum:** ligate the artery in the interaortocaval space, so the kidney decompresses; then double-ligate the flaccid vein.
4. Alternative: start at the lower pole and lift the kidney from posterior to anterior to show the artery.
5. **Left:** mobilise the splenic flexure down to the iliacs; divide the renocolic ligament (pad the pancreas).
6. Follow the aorta or gonadal vein to the LRV. Tie the artery above and behind it, confirm occlusion, then the vein.

KEY POINT

Why artery first? Tying the vein first leaves the kidney engorged, so collaterals bleed. Arterial ligation shrinks the kidney and softens the vein, which can then be checked for thrombus. A persisting pulse means an accessory artery, incomplete control, or the SMA has been tied.

- **Lumbar veins:** use silk ties and **no clips on the IVC side**, because clips block emergency caval clamping.
- **Lost pedicle control:** tell the anaesthetist, compress, use two suckers and **never clamp blindly**.
- **Adrenalectomy** only for an upper-pole tumour with a lost plane, an adrenal mass or diffuse involvement. Metastasis occurs in **<5%**, and contralateral metastasis is as common as ipsilateral.
- **LND:** nodes are positive in ≈5%, and the EORTC RCT showed no overall survival benefit, though it included many low-stage tumours.
- **Do LND if nodes are enlarged**: a short axis **≥1.5 cm ≈ 50% positive**. Consider it in high risk (>10 cm, grade ≥3, sarcomatoid, thrombus, T4).
- The template runs from crus to bifurcation. **Clip lymphatics above the renal vein** to prevent chylous ascites.
- **Closure:** fill with saline and give a big breath; bubbles mean a pleural tear, which is sutured over a 14-Fr catheter. Take a **chest X-ray in all**.

5 · IVC TUMOUR THROMBECTOMY

Surgery is the only cure; thrombus level decides the plan.

- RCC is the commonest cause (**5–10%** of RCC). Untreated, most die within 1 yr, so surgery (RN and LND, with or without systemic therapy) is the backbone.
- **Tumour thrombus** differs from **bland thrombus** (clot below the obstruction). **MR venography** best shows extent, wall invasion and bland thrombus.
- Anticoagulate for venous thromboembolism, bland thrombus or IVC occlusion. Use **no suprarenal filters**, because thrombus grows onto the device.
- Use **transoesophageal echo (TEE)** for levels II–IV, and involve a **cardiothoracic surgeon for III–IV**.

Mayo level	Upper extent	Management
0	Renal vein	RN (commonest)
I	IVC **≤2 cm** above ostium	Milk back, Satinsky partial clamp, ostial cavotomy
II	>2 cm, below hepatic veins	Full IVC control; infrahepatic cavotomy
III	At or above major hepatic veins, below diaphragm	Liver mobilisation with or without Pringle; occlusive thrombus: suprahepatic clamp; nonocclusive: venovenous bypass (VVB)
IV	Above diaphragm	Sternotomy, cardiopulmonary bypass (CPB) with deep hypothermic circulatory arrest (DHCA), atriotomy

Bland group	Location	Action
A	None	None
B	Iliac veins	Infrarenal filter
C	Infrarenal IVC, separate	Interrupt (clip or staple); do not extract
D	Mixed with tumour	Resect infrarenal IVC; keep lumbars as collaterals

ALGORITHM

1. **Level I (right):** control the IVC before handling the tumour; ligate the right artery interaortocavally.
2. Loop the LRV and the suprarenal and infrarenal IVC with **Rummel tourniquets**.
3. Milk the thrombus toward the ostium; place a Satinsky round the ostium with the thrombus inside the jaws.
4. Do an ostial cavotomy keeping ≥half the IVC width, and extract.
5. **Level I–II:** ligate the renal artery early. Tighten tourniquets in order: **infrarenal IVC, contralateral renal vein, suprarenal IVC**.
6. Open the cava, extract, flush and inspect the wall. Close with 4-0 Prolene.
7. Release the **caudal clamp first**, flush on a held breath, then release the cranial clamp.

- A **left tumour (level II)** is harder: tie the left artery first, then Kocher, **split-and-roll** the IVC and excise the LRV ostium en bloc. Do not clamp the right renal artery.
- **Level III:** keep bypass on standby and mobilise the liver as in transplantation. The **major hepatic veins are never divided**, and **Budd–Chiari syndrome is a contraindication**.
- The **Pringle** manoeuvre is needed if clamping above the hepatic veins. Limit it to ≤60 min, ideally **≤20 min**.
- **Level III sequence:** clamp infrarenal IVC, contralateral renal vein, porta hepatis, then suprahepatic IVC; open and extract. Unscrapable wall invasion needs IVC resection.
- Do a 2–5 min **clamp trial**. If cardiac output falls by >50% or mean arterial pressure by >30%, use bypass. Occlusive thrombus tolerates clamping because of azygos collaterals; free-floating thrombus does not.
- Incomplete tumour removal is universally fatal, so never compromise IVC control to save time.

	VVB	CPB with DHCA
Access	Femoral to jugular, percutaneous	Sternotomy; cool to 16–18 °C and drain 95% of blood
Heparin	Partial	Full; coagulopathy
Drawback	Lumbar and intercostal back-bleeding	Cardiac and neurological events
Use	Least invasive; not with atrial or iliac bland thrombus, Budd–Chiari	Level IV; bloodless field; ≤60 min (<40 better)

- In **VVB**, place cannula tips far from the thrombus, because dislodgement causes massive pulmonary embolism (PE).

P3 *Vascular grafts and vascular surgery in urology — caval repair: ≥50% lumen primary closure, else pericardial patch (Leibovich–Bower width), circumferential: PTFE 16–20 mm + lifelong aspirin 81 mg; left tumour needs right-kidney drainage*

Not in the paper answer:

- Suprarenal IVC ligation with a right tumour is acceptable if the LRV is tied caval to its tributaries, which act as collaterals.

KEY POINT

"Shortcuts" that become "longcuts": opening the IVC before all inflow is controlled; side-biting a level I; clips on the IVC side; incomplete arterial control; forceful milking (embolus, tear); clamping before volume loading or skipping the test clamp.

6 · PARTIAL NEPHRECTOMY

P2 *Nephron-sparing surgery / partial nephrectomy in RCC: role, indications and approaches — PN standard for T1; imperative/relative/elective; relative contraindications (<20% nephron mass, pedicle encasement, thrombus, cT4, nodes); hereditary observe to 3 cm except HLRCC*

Not in the paper answer:

- Hyperfiltration injury (focal segmental glomerulosclerosis, proteinuria) is rare unless nephron mass falls by >80%.
- Multifocality is ≈2% in clear cell and ≈10% in papillary RCC, so inspect the whole kidney with intraoperative ultrasound.
- PN is size-independent if it is safe; collecting-system invasion is a relative contraindication.

Question	Evidence
Warm ischaemia	Aim **<30 min**; within limits, preserved parenchyma dominates
Cold vs none	Randomised controlled trial (RCT): no difference
Mannitol	RCT: no benefit
On- vs off-clamp	2 RCTs: no difference
3D models	RCT: less ischaemia, blood loss and stay

> **KEY POINT**
> PN trades nephrons lost to the margin against nephrons injured by ischaemia. Within accepted limits, the **volume of parenchyma preserved** matters more than clamp minutes. Hypothermia and mannitol are optional.

- Always **control the hilum first**, even if not clamping, so a clamp can go on at once.
- **Enucleation** follows the pseudocapsule plane; **resection** takes a 5–10 mm rim and is usually clamped.
- **Renorrhaphy:** the inner layer closes vessels and collecting system with absorbable sutures (permanent sutures become a stone nidus); the outer layer is capsular over a bolster.

7 · COMPLICATIONS & MANAGEMENT

Injury	Management
SMA	Injured when mistaken for the left renal artery or by Cattell-Braasch traction. Potentially lethal, so call a vascular surgeon for primary repair or graft
Inferior mesenteric artery or vein (IMA/IMV)	Ligation safe if the marginal artery of Drummond is patent
Superior mesenteric vein (SMV)	Ligation causes bowel ischaemia, splanchnic hypertension and hypotension. **Never ligate**; repair with 6-0 Prolene venorrhaphy. If transected, reanastomose or use autologous vein; leave the abdomen open
Duodenum	Haematoma narrowing the lumen: incise serosa and muscularis, drain. Tear: debride, 2-layer closure, omentum, drain, nasogastric tube
Pancreas	Glue and drain (check amylase); ductal injury: general surgeon; late collection: percutaneous drain, nil by mouth, octreotide
Pleura	Purse-string over a 14-Fr catheter, evacuate on a positive-pressure breath; large tear: chest tube

- In thrombectomy, air embolism is prevented by releasing the caudal clamp first. Use **no cell saver**, because it spreads tumour.
- A **urine leak** mostly appears within 72 h. If drain output is >50 mL/24 h, a drain creatinine above serum means urine; a persisting urinoma needs a double-J stent and Foley, and most resolve in 4–6 wk.
- A significant **pseudoaneurysm** (flank pain, haematuria or shock) needs **angiography** with **selective embolisation**, because CT angiography can miss it and cannot treat it.
- An **arteriovenous fistula** is observed, then coil-embolised, with surgery if complex.
- **Renal insufficiency** is usually mild and transient; exclude obstruction, drug toxicity and thrombosis.

142 Laparoscopic and Robotic Surgery of the Kidney

1 · PRINCIPLES & PATIENT SELECTION

Fitness here means fitness for the gas.

- First laparoscopic nephrectomy: **Clayman, 1991** MCQ; first laparoscopic partial nephrectomy (PN): Winfield, 1993.
- Minimally invasive surgery (MIS) gives less pain, blood loss and convalescence than open, with **equivalent long-term outcomes** and the same indications.
- **General anaesthesia is mandatory.** The CO_2 pneumoperitoneum impairs ventilation, venous return and cardiac output, so risk is high in chronic obstructive pulmonary disease (COPD) and cardiac disease.
- In severe COPD with hypercarbia, lower the pressure, then use helium, then convert.
- With the robot, **docking and undocking delay emergency access**, so the team must communicate closely.
- In obesity, **visceral adipose tissue**, not body mass index (BMI), predicts difficulty. Prior surgery is **not a contraindication**.

> **KEY POINT**
> The patient must tolerate the **pneumoperitoneum**, not the renal operation: cardiopulmonary reserve, not age or BMI, decides fitness.

2 · ACCESS & PORT PLACEMENT

Routes are transperitoneal (TP, most used), retroperitoneal (RP), hand-assisted (HAL) and single-site (LESS).

	Transperitoneal	Retroperitoneal
Space / landmarks	Large; familiar	Limited, poor triangulation; less familiar
Bowel	Colon reflection; adhesiolysis	Peritoneum not entered
Hilum	Reached after colon reflection	Direct, especially the posterior aspect
Preferred for	Most surgery; suturing; anterior tumours	Posterior tumours, pyeloplasty, cysts, biopsy; dense adhesions, peritoneal dialysis, pregnancy, morbid obesity

- **TP** uses a 30–45° modified flank position with 3–5 trocars.
- **Robotic** arms are spaced **8 cm (Si) / 6 cm (Xi)**, forming a **broad-based triangle** with the camera at the apex.
- **RP** uses the full flank position; the space is made by finger or balloon below the 12th rib in the posterior axillary line.

> **KEY POINT**
> RP pitfalls: a balloon inflated within muscle causes a flank hernia; too anterior injures the peritoneum or colon; too posterior makes the quadratus or psoas bleed. Bail-out: open the peritoneum and continue transperitoneally.

- **HAL** gives tactile feedback and helps novices, donors and scarred kidneys; a hand through the port can compress a bleeding vessel.
- **LESS and the single-port robot** improve cosmesis but lose triangulation, with unchanged outcomes.

3 · NEPHRECTOMY — OPERATIVE STEPS

ALGORITHM

1. Reflect colon; find ureter and gonadal vessels on the psoas.
2. Lift kidney to expose the hilum; artery, then vein.
3. Upper pole (adrenal spared or taken); ureter and lateral attachments; bag; extract.

- Keep the **lateral attachments** until late, or the kidney rolls over the hilum. On the right, dissect cold near the duodenum and do a **Kocher**.
- The **ureter** lies just behind the gonadal vessels and is a useful handle.
- **Hilum:** divide the **artery first, then the vein**, with an endo-GIA stapler. **Never leave clips in the staple line.**
- **En bloc hilar stapling** is safe (no arteriovenous fistula, AVF) and saves ~45 min.
- **Extraction:** intact in a bag, or morcellated in an impermeable **LapSac**, which keeps type and grade but loses size and stage.

4 · SIMPLE VS RADICAL NEPHRECTOMY

	Simple	Radical
Gerota's fascia and fat	May be left	Removed intact with specimen
Pathology	Benign, often inflammatory	Renal cell carcinoma (RCC)
Difficulty / conversion	Can exceed radical; higher	Planes preserved; lower

- **Simple** nephrectomy is for benign nonfunctioning kidneys, such as chronic pyelonephritis, xanthogranulomatous pyelonephritis, obstruction, reflux or tuberculosis; after pyonephrosis or stones, use hand assistance or a subcapsular approach.
- **Radical nephrectomy (RN):** MIS equals open oncologically at 5 and 10 yr. In RCTs of RP versus TP, the only consistent RP advantage is **shorter operative time**.
- **Lymph node dissection (LND)** has staging value only, with no overall survival benefit; the AUA advises removing clinically positive nodes.
- Tumours >10 cm are feasible, though landmarks are lost; conversion is higher laparoscopic than robotic.
- Selected **inferior vena cava (IVC) thrombus** cases can be done by experts, with early arterial ligation that may retract the thrombus (Ch 141). Laparoscopic **cytoreductive** nephrectomy allows earlier systemic therapy.

KEY POINT

Access changes morbidity and recovery, never oncology: early arterial control, en bloc removal and intact bagged extraction carry over from open surgery.

5 · PARTIAL NEPHRECTOMY

P2 P1 *Nephron-sparing surgery / partial nephrectomy in RCC; Role of renal nephrometry in management of RCC — PN default where feasible; RENAL score predicts complexity, WIT, conversion; enucleation; TP anterior vs RP posterior*

Not in the paper answer:

- In cT1b/T2 tumours, PN gives better estimated GFR with equal oncology, but **tumours >5 cm upstaged to pT3a recur more after PN**.
- The collecting system is entered in about 60%, so always drain.

ALGORITHM

1. Intraoperative ultrasound (size, depth); open Gerota's away from the lesion; score the capsule.
2. Clamp the hilum (Satinsky en bloc or bulldogs); cold-scissor excision.
3. Close vessels and collecting system; renorrhaphy; unclamp; drain.

- **Trifecta:** negative margin, no major complication within 30 days, and **WIT ≤25 min**.
- **Enucleation** gives less blood and function loss but more positive margins, with the same survival. It suits familial, multifocal, solitary-kidney and severe chronic kidney disease (CKD) cases.
- **Renorrhaphy:** sliding-clip and barbed sutures shorten warm ischaemia time (WIT), and **single-layer** closure gives better function and fewer pseudoaneurysms.
- The **collecting system** is entered in **~60%**; repair it with absorbable suture and **always drain**.
- **Ischaemia:** the historical warm limit is **30 min**, but **preserved parenchymal volume predicts function better than WIT**. Use cold ischaemia for a solitary kidney, CKD or a long expected clamp time.

Hilar strategy	Best for	Verdict
Artery-only vs artery and vein	Any	Equivalent
Early unclamping	Any	Possibly fewer pseudoaneurysms
Selective / super-selective (near-infrared fluorescence, duplex)	Mapped segmental supply	Randomised: no 6-month benefit
Off-clamp (zero ischaemia)	Small, exophytic	Better early eGFR; no 2-yr difference; more blood loss
Parenchymal compression	Peripheral	Feasible, safe
Hypothermia	Solitary kidney, long WIT	Benefit unproven

KEY POINT

No clamping strategy has beaten standard clamping on long-term function: what is kept matters more than how long it is clamped.

- Robotic PN has fewer conversions and shorter WIT than laparoscopic, and less blood loss than open.
- **Positive margins (0–7%)** matter only in high-risk tumours, so the answer is **surveillance, not reoperation**.
- **Port-site recurrence is ~0.1%** and follows spillage, not the gas.
- A **hilar** tumour touches the renal artery or vein; consider enucleation with superficial renorrhaphy bites.
- Salvage surgery after PN or ablation often ends in RN, because the planes are lost.

6 · OTHER RENAL PROCEDURES

Procedure	Indication	Key points
Cyst decortication	Pain, infection, obstruction, suspected malignancy	Percutaneous aspiration first proves the cyst causes the pain; excise the wall, biopsy, fulgurate the base
Laparoscopic renal biopsy	Failed percutaneous biopsy, bleeding diathesis, solitary kidney, morbid obesity	RP, two ports, lower-pole cortex
Nephropexy	Descent >5 cm or 2 vertebrae from supine to erect, with pain, reduced flow or obstruction	Usually right side, slim young women; 20% asymptomatic. Fix the capsule to psoas or quadratus fascia

Procedure	Indication	Key points
Calyceal diverticulectomy	Symptomatic; endoscopy failed	Suture the neck, obliterate the cavity with fat
Nephrolysis	Chyluria, loin pain–haematuria	Lateralise by cystoscopy; hilar lymphatic ligation with or without omental wrap
Ablation	≤3 cm, comorbid, solitary kidney, hereditary RCC	Percutaneous preferred. Cryo: two freeze–thaw cycles, iceball 0.5–1 cm beyond tumour. Radiofrequency ablation (RFA): 60–100°C (>105°C vaporises), 1 cm margin, no real-time imaging; more retreatment

7 · COMPLICATIONS & MANAGEMENT

Dangerous complications happen out of view, so seek them.

P1 *Pneumoperitoneum physiology, complications of retroperitoneoscopy and laparoscopic donor nephrectomy — vascular injury commonest (<1/1000, fatal 8–17%); venous bleed control; gas embolism (left side down, head down, aspirate); pneumothorax; thermal bowel injury, CT with oral contrast; rhabdomyolysis*

Not in the paper answer:

- The overall complication rate is **~15%** (PN > RN > simple), with a learning curve of about 20 nephrectomies.
- **In left RN, the superior mesenteric artery (SMA) can be mistaken for the renal artery**, leading to bowel necrosis.

ALGORITHM

1. Major bleed: locate; temporary control (grasper or pressure); inform the anaesthetist (blood, open set).
2. Clear the field: raise pneumoperitoneum or add a trocar; then definitive clip or suture.
3. If that fails, then convert and call a vascular surgeon.

- Vascular injury happens at **access** or at the hilum. **Never clip or suture blindly.**
- At the end of the case, **lower the insufflation pressure** to unmask venous bleeding.
- **Haematuria at ~day 12 after PN** suggests a pseudoaneurysm or AVF and is treated by selective embolisation.
- A **urine leak** after PN (0.8–10%) is drained percutaneously, with a stent if it persists.
- **Bowel injury** (<1%, mortality ~3%) is thermal in about half. It presents with **pain at a single trocar site**, distension, diarrhoea and a **normal or low white count**.
- Pancreatic injury is insidious, so inspect at the end. A diaphragm injury shows as a sudden rise in airway pressure and is closed with a purse-string, followed by a chest X-ray.
- Bowel injury is repaired laparoscopically if stable, or by laparotomy if unstable or with gross spillage.
- **Neuromuscular** injury (2.7%) includes flank rhabdomyolysis, treated by hydration.

KEY POINT

MIS complications are the open ones in altered form and timing: the lethal ones (access vessel injury, thermal bowel injury) happen out of view and present late or atypically.

8 · LAPAROSCOPIC LIVE DONOR NEPHRECTOMY †

P4 P1 *Laparoscopic donor nephrectomy — left preferred (longer vein), donor keeps better kidney; gonadal–ureter packet; artery then vein; no Hem-o-lok on donor artery; mortality 3/10 000; lifetime ESRD risk*

Not in the paper answer:

- † Donors need lifelong annual blood pressure and renal function checks.

143 Focal Therapy for Renal Tumours

1 · RATIONALE & INDICATIONS

Ablation suits patients whom surgery harms more than the tumour.

- Small renal cell carcinoma (RCC) rarely kills; chronic kidney disease (CKD) and comorbidity do (Ch 137).
- **Partial nephrectomy (PN) is the gold standard** for cT1a. Thermal ablation (TA) is the alternative: cryoablation (CA), radiofrequency ablation (RFA) or microwave ablation (MWA).
- **Indications:** high surgical risk, solitary kidney, bilateral tumours, or hereditary syndromes (von Hippel–Lindau, VHL) needing repeat treatment. Recovery is faster and renal function equal or better than PN.

KEY POINT

Trap: TA is never the default for a fit patient.

Guideline	Position on TA
American Urological Association (AUA) 2021	Consider for cT1a **≤3 cm**, especially comorbid; PN prioritised; percutaneous preferred
European Association of Urology (EAU) 2022	For frail/comorbid; TA not routinely **>3 cm**; CA not **>4 cm**
National Comprehensive Cancer Network (NCCN) 2022	**Only one allowing cT1b** (non-surgical candidates); may need multiple treatments
American Society of Clinical Oncology (ASCO) 2017	Option if complete ablation achievable

2 · SELECTION & WORK-UP

The tumour stays in place, so biopsy gives the only histology.

ALGORITHM

1. Counsel: age, comorbidity, CKD risk, size and location.
2. Biopsy, then re-counsel with histology.
3. Choose PN, surveillance or percutaneous ablation.

- **The AUA advises biopsy of all**, ideally at a **separate earlier session**. Post-ablation, biopsy only a new enhancing lesion.
- Treat urinary infection first; colonised urine (ileal conduit) risks abscess.
- Complication predictors are the RENAL score (low 4–6, moderate 7–9, high 10–12), American Society of Anesthesiologists (ASA) score >3 (about 3×) and $(MC)^2$ (CA only). Day-case suits ≤3 cm, low complexity and no anticoagulation.

3 · MECHANISMS

P2 *Biological effects of cryosurgery — freeze/thaw cell injury, vascular thrombosis, apoptosis, immunomodulation; argon cools/helium warms; kidney dies at −19.4 °C; <−20 °C only 3 mm inside iceball edge*

Not in the paper answer:

- Cryo target is **≤−40 °C**; the iceball extends **5–10 mm beyond tumour**, pushing the −40 °C isotherm past the tumour edge.
- Heat: **≥60 °C** causes coagulative necrosis; ≥100 °C causes vaporisation and charring. Lethal: 45 °C for 60 min, 55 °C for 5 min, **70 °C for 1 min**.

- **Heat sink:** large vessels carry heat or cold away, so perivascular tumour survives. Counter with colder or longer freezes, more probes, or MWA.

4 · CHOICE OF ENERGY

- RFA electrodes vary: the cool-tip gives the largest zone, the umbrella the most spherical, and a single monopolar tip suits only ≤2 cm.

	Cryoablation	RFA	MWA
Physics	Argon cools (**Joule–Thomson**); helium warms	450–1200 kHz current causes ionic friction	915 MHz/2.45 GHz causes **dielectric hysteresis**
Target	≤−40 °C	≥70 °C; set 90–105 °C	Often >100 °C
Cycles	**Double freeze–thaw** (10 + 6–8 min)	Two, 30-s cool-down	Single dose (e.g. 65 W × 5 min)
Margin	Iceball 5–10 mm	≥5 mm	≥5 mm
Charring limits	No	**Yes**	No
Grounding pad	No	Yes	No
Heat sink	Yes	Yes	**Least**
Zone visible	**Iceball**	**No**; inferred from impedance and temperature	Gas bubbles on ultrasound (US)
Pros	Spares collecting system	Longest follow-up	Fast; fewest complications
Cons	Bleeding, tumour fracture	Heat sink, pad burns	Risk to adjacent organs

KEY POINT

Why RFA is self-limiting: charring raises impedance and halts current (cap 105 °C, electrodes cooled). MWA radiates energy, so it works through desiccated tissue and blood, heats actively to 2 cm (RFA: millimetres) and outruns the heat sink.

- In CA the second freeze enlarges necrosis. MWA for cT1 uses 45–65 W for 5–15 min.

5 · TECHNIQUE

- **Percutaneous is standard.** Laparoscopic is kept for prior scarring or failed hydrodisplacement. General anaesthesia with jet ventilation reduces renal motion.

ALGORITHM

1. Give intravenous (IV) antibiotics.
2. Position prone or flank on the computed tomography (CT) gantry.
3. Place probes under image guidance; confirm on non-contrast CT.
4. Ablate, then contrast CT for completeness and haematoma.

- **Hydrodisplacement** pushes organs away (colon >1 cm) with saline or 5% dextrose (**D5W for RFA**, as saline conducts current). Use caution in cardiac failure.
- **Pyeloperfusion** (central tumours) irrigates a ureteral catheter at the **opposite temperature**: cold for RFA/MWA, warm for CA.
- **CA:** cryoprobes sit **1–2 cm apart** and come out only after full thaw. Bleeding rises with size and probe number (>4 cm). **RFA:** ≥70 °C in each quadrant during cool-down excludes heat sink; overlap ablations for larger lesions.
- **MWA:** antennas go tangentially, never into the collecting system: 1 for ≤2.5 cm, 2 for 2.5–3.5, 3 for 3.5–4.5 cm.

6 · FOLLOW-UP, SUCCESS & SALVAGE

With no specimen, imaging alone shows success.

- **Success is complete loss of contrast enhancement** on CT/MRI.
- **Incomplete ablation** enhances on the first scan and is re-ablated. **Local recurrence** later enlarges or enhances and needs biopsy. **Enlargement after any modality is recurrence until proved otherwise.**
- CA zones shrink >50% in year 1; RFA zones barely shrink (**fibrotic halo**).
- **AUA:** CT/MRI with and without contrast plus chest imaging at **6, 12, 24, 36, 48, 60 months**, then shared decision. US is not routine unless a contrast-enhanced (CEUS) protocol exists.
- Salvage is re-ablation, PN, radical nephrectomy (RN), or stereotactic body radiotherapy (SBRT) if comorbid.

7 · ONCOLOGICAL OUTCOMES VS PN

- There is no randomised trial. Local recurrence is commoner after TA but **equal once re-ablation is counted**.
- **Metastasis-free and cancer-specific survival are equal**; overall survival is lower in these older, comorbid patients.
- **Size predicts failure:** RFA does worse at ≥3 cm, and T1b worse than T1a. No energy is superior.

> **KEY POINT**
>
> **Why comparisons are biased:** benign and under-graded tumours flatter TA; occult pT3a penalises it; non-surgical candidates die of other causes first. Few RCC deaths occur, so trials need huge cohorts.

8 · COMPLICATIONS & MANAGEMENT

- Major complications (Clavien ≥3) are **≤5%**, fewer than PN; percutaneous beats laparoscopic, and MWA is lowest.

Complication	Prevention	Management
Haemorrhage – commonest; transfusion 2–3%	Full thaw before CA probe removal; avoid intercostal arteries	Ablate (RFA/MWA coagulate); otherwise selective embolisation
Urinoma, fistula, pelviureteric junction (PUJ) or ureteric stricture	Pyeloperfusion; CA for central tumours	Conservative or stent; drain a large urinoma
Bowel or other organ injury (tumour <1 cm away)	Hydrodisplacement, positioning	Fistula: stent, low-residue diet; diversion if persistent
Pneumothorax (access above 12th rib)	Recognise during procedure	Aspirate; chest tube rarely
Nerve injury / RFA pad burn	Pads level on posterior thighs	Neuralgia is self-limiting
Infection <1%, may be lethal; abscess at 1–6 months	Urine culture, prophylaxis	†Drain and antibiotics

9 · NEWER MODALITIES (INVESTIGATIONAL)

- **High-intensity focused ultrasound (HIFU)** ablates incompletely; **histotripsy** (non-thermal) has no clinical data yet.
- **SBRT** needs no anaesthesia but leaves viable tumour on biopsy; it may have a salvage role.
- **Irreversible electroporation** spares vessels but controls cancer poorly.

144 Treatment of Advanced Renal Cell Carcinoma

1 · OVERVIEW

In metastatic RCC, drug choice rests on risk group and mechanism.

P2 *Metastatic RCC (mRCC): assessment and management, including recent concepts — work-up, MSKCC/IMDC, first-line by risk, switch-mechanism later lines, CARMENA/SURTIME, metastasectomy 35–50% 5-yr, KEYNOTE-564, nccRCC*

Not in the paper answer:

- About **1/3** present with metastases; a further **20–40%** of localised tumours metastasise later. mRCC 5-yr survival is about **12%**; 10-yr <5%.

Standard care is now **immune checkpoint (immuno-oncology, IO) combinations**.

2 · PROGNOSTIC MODELS

- **MSKCC (Motzer)** came from the cytokine era; **IMDC (Heng)** came from first-line VEGF-targeted therapy.

Factor	MSKCC (5)	IMDC (6)
Performance status	Karnofsky <80%	Karnofsky <80%
Haemoglobin	< lower limit of normal (LLN)	< LLN
Corrected calcium	**>10 mg/dL**	> upper limit of normal (ULN)
Lactate dehydrogenase (LDH)	**>1.5 × ULN**	—
Time factor	No prior nephrectomy	**<1 yr** from diagnosis to systemic therapy
Neutrophils; platelets	—	> ULN (each a factor)
Favourable (0) – median overall survival (OS)	20 mo	43.2 mo
Intermediate (1–2)	10 mo	22.5 mo
Poor (≥3)	4 mo	7.8 mo

KEY POINT

Three factors are shared. IMDC swaps LDH and prior nephrectomy for neutrophilia, thrombocytosis and short interval. The IMDC risk group **decides the first-line regimen**.

3 · BIOLOGY AND DRUG TARGETS

- The **VHL** protein (3p) tags **hypoxia-inducible factor (HIF)-α** for degradation; when VHL is lost, HIF drives VEGF, PDGF, TGF-α, GLUT-1 and erythropoietin. Drugs block the ligand (bevacizumab), receptors (tyrosine kinase inhibitors, TKIs), HIF translation (mTOR inhibitors) or HIF-2α (belzutifan). MET/AXL/FGFR escape explains broader later TKIs.
- **CTLA-4** and **PD-1/PD-L1** switch off T cells. Blocking them gives autoimmune toxicity, and benefit shows as OS and durability rather than progression-free survival (PFS).
- VEGF inhibitors are immunomodulatory, the rationale for IO plus TKI. Clear cell RCC (ccRCC) is chemoresistant.

Agent	Target / route	Signature toxicity / note
Bevacizumab	Antibody vs VEGF-A; IV	Bleeding, proteinuria, hypertension (HTN)
Sorafenib	VEGFR2, PDGFR-β, raf-1	Rash, hand–foot syndrome (HFS); rarely used
Sunitinib	VEGFR2, PDGFR-β, c-KIT, FLT3; **50 mg, 4 wk on / 2 off**	HFS, fatigue, HTN, thrombocytopenia, **hypothyroidism**

Agent	Target / route	Signature toxicity / note
Pazopanib	VEGFR, PDGFR, KIT	**Hepatotoxicity (raised ALT)**; less fatigue/HFS
Axitinib	VEGFR1–3 only – most selective	Diarrhoea, fatigue, HTN
Tivozanib	VEGFR1–3, c-KIT, PDGFR-β	Third line
Cabozantinib	VEGFR, **MET, AXL**	Fatigue, diarrhoea, HFS; frequent dose reduction
Lenvatinib	VEGFR1–3, **FGFR1–4**, PDGFR-α, RET, KIT	Partner of everolimus or pembrolizumab
Temsirolimus	mTOR; **IV weekly**	Mucositis, hyperglycaemia, hyperlipidaemia, pneumonitis
Everolimus	mTOR; oral daily	Asthenia, anaemia
Belzutifan	**HIF-2α inhibitor**; oral	After IO and TKI; also VHL-associated RCC
Nivolumab, pembrolizumab	Anti-PD-1; IV	Immune-related: colitis, hypophysitis, vitiligo
Avelumab	**Anti-PD-L1**; IV	
Ipilimumab	Anti-CTLA-4; IV	

4 · SURGERY IN METASTATIC DISEASE

Surgery now serves systemic therapy; the question is who and when.

- **Cytoreductive nephrectomy (CN) rationale:** spontaneous regression of metastases is <1%, and a bulky primary suppresses T cells. CN helps only as a platform for systemic therapy.
- Fewer than 1/4 reach systemic therapy with ECOG 2, multiple sites, or brain or liver metastases.
- **CARMENA caveats:** about 45% were poor risk, about 25% of the CN arm were off-protocol, and it stopped early underpowered. It does **not** show harm in fit, low-burden, favourable patients.

ALGORITHM

1. Assess mRCC: histology, IMDC risk, PS and metastatic burden.
2. If fit, with low burden, favourable features and a resectable primary, then consider upfront CN followed by systemic therapy.
3. If intermediate/poor risk or high burden, then give systemic therapy first and offer deferred CN to responders.

KEY POINT

CN helps only patients fit enough, with disease slow enough, to reach systemic therapy; operating first on poor-risk patients delays it.

- **Metastasectomy** does best with complete resection, a solitary lesion, age <60, disease-free interval >1 yr, lung site, size <4 cm and metachronous disease, all markers of indolent biology.
- **Palliative** nephrectomy treats pain and paraneoplastic syndromes; try **angioembolisation for haematuria** first. Brain and spinal metastases get surgery and radiation.

5 · SYSTEMIC THERAPY FOR CCRCC

First line follows IMDC risk; later lines switch mechanism.

	IFN-α	High-dose IL-2
Response	10–15%	15–20%
Durable complete response (CR)	<2%	**7–9%**; >60% of CRs stay disease-free

	IFN-α	High-dose IL-2
Regimen	Schedule/route immaterial	**600,000–720,000 IU/kg** IV bolus 8-hourly, max 15 doses
Toxicity	Constitutional	**Vascular leak**: hypotension, multiorgan failure; early mortality 2–5%

- IL-2 is now kept for carefully selected fit patients with clear cell histology and prior nephrectomy; IFN-α is not used first line.

Risk	Preferred / standard	Alternatives (incl. if IO not possible)	Certain circumstances
Favourable	Pembrolizumab + axitinib; nivolumab + cabozantinib; pembrolizumab + lenvatinib	Avelumab + axitinib, cabozantinib, ipilimumab + nivolumab, pazopanib, sunitinib	**Active surveillance**, axitinib, high-dose IL-2
Intermediate / poor	Same three **+ ipilimumab + nivolumab**	Cabozantinib, avelumab + axitinib, pazopanib, sunitinib	Axitinib, high-dose IL-2, **temsirolimus**

KEY POINT

Choosing: IO/TKI gives higher response and PFS and is better tolerated; IO/IO gives deeper durable responses (CR 9%) with autoimmune toxicity; never compared directly. Ipilimumab + nivolumab is restricted to intermediate/poor risk (favourable did better on sunitinib in CheckMate 214). PD-L1 does not select. Use a TKI or temsirolimus if IO is contraindicated.

Prior therapy	EAU standard	NCCN recommended (no preferred)
TKI only (IO-naive)	**Nivolumab or cabozantinib** (alternative axitinib)	IO combinations, nivolumab, cabozantinib, lenvatinib + everolimus
Prior IO	Any VEGF-targeted agent not previously used	Axitinib, **belzutifan**, cabozantinib, lenvatinib + everolimus, **tivozanib**

- Tivozanib is used after ≥2 prior therapies. TKI + mTOR and nivolumab + sunitinib/pazopanib combinations failed.
- **Ipilimumab + nivolumab toxicity:** grade 3–4 events **46%**, high-dose steroids 35%, discontinuation 22%. Dose: nivolumab 3 mg/kg + ipilimumab 1 mg/kg 3-weekly × 4.
- † For immune-related events, hold IO and give steroids for grade ≥2 (prednisolone 1–2 mg/kg for grade 3–4), tapering over ≥4 weeks. Grade 4 means permanent stopping.
- TKI toxicity is managed by dose reduction or interruption; mTOR inhibitors need glucose, lipid and phosphate monitoring.

6 · NON-CLEAR CELL RCC

- Non-clear cell RCC (nccRCC) is 15–25%. No agent is proven, so offer a **trial to all**.
- Papillary tumours respond to sunitinib in only 5–10%; temsirolimus beat IFN in the nccRCC subgroup.
- **Hereditary papillary RCC (HPRC)** carries a germline activating **MET** mutation (chromosome 7) and gives type 1 papillary RCC. Sporadic papillary has **trisomy 7 in >2/3**. MET inhibitors (savolitinib, crizotinib) are active here.
- **Hereditary leiomyomatosis and RCC (HLRCC)**: fumarate hydratase loss gives type 2 papillary RCC. The cells depend on glycolysis, so **bevacizumab + erlotinib** restricts glucose (partial response about 2/3).
- **Collecting duct** carcinoma is treated with **gemcitabine + carboplatin**; **sarcomatoid** tumours with gemcitabine-based therapy.

7 · LANDMARK TRIALS

Trial	Design	Result	Take-home
SWOG 8949 / EORTC	CN + IFN vs IFN	OS 11.1 vs 8.1; 17 vs 7 mo; response same	CN in cytokine era
CARMENA	CN + sunitinib vs sunitinib; noninferiority; n=450	OS 13.9 vs **18.4** mo, hazard ratio (HR) 0.89	No upfront CN in intermediate/poor
SURTIME	Immediate vs deferred CN	OS 15.0 vs **32.4** mo, HR 0.57	Systemic therapy first is safe
Sunitinib vs IFN	Phase III, first line	PFS 11 vs 5 mo; response 31 vs 6%	Later comparator
COMPARZ	Pazopanib vs sunitinib; noninferiority	PFS HR 1.05; OS similar	Equivalent; patients preferred pazopanib (PISCES)
CALGB 90206 / AVOREN	Bevacizumab + IFN vs IFN	PFS better, OS not	Little used
Temsirolimus phase III	vs IFN vs combination; poor risk, all histologies	OS 10.9 vs 7.3 mo	Only mTOR inhibitor with OS gain
TARGET	Sorafenib vs placebo after cytokines	PFS 5.5 vs 2.8 mo; OS same	Historical
Everolimus phase III	vs placebo after sunitinib/sorafenib	PFS 4.0 vs 1.9 mo; OS same	Supplanted as monotherapy
Lenvatinib ± everolimus	Phase II, after VEGF therapy	Combination PFS longer	Lenvatinib + everolimus
AXIS	Axitinib vs sorafenib, second line	PFS 6.7 vs 4.7; OS same	Second-line axitinib
METEOR	Cabozantinib vs everolimus after VEGF	PFS HR 0.58; **OS HR 0.70**	Only TKI with OS gain over active comparator
CABOSUN	Phase II; cabozantinib vs sunitinib; intermediate/poor	PFS 8.2 vs 5.6, HR 0.66	First-line cabozantinib
TIVO-3	Tivozanib vs sorafenib, after 2–3 lines	PFS HR 0.73; OS same	Third-line tivozanib
LITESPARK-005	Belzutifan vs everolimus after IO + TKI	PFS HR 0.75 (curves separate late)	HIF-2α inhibition approved
CheckMate 025	Nivolumab vs everolimus after TKI; primary OS	OS 25.0 vs 19.6, **HR 0.73**; PFS same	IO benefit shows as OS
CheckMate 214	Ipilimumab + nivolumab vs sunitinib	Intermediate/poor OS HR 0.63; CR 9 vs 1%	Only IO/IO; not for favourable
KEYNOTE-426	Pembrolizumab + axitinib vs sunitinib	PFS and OS better; response 60 vs 40%	All risk groups
JAVELIN Renal 101	Avelumab + axitinib vs sunitinib	PFS better; OS not significant	Not NCCN-preferred
CheckMate 9ER	Nivolumab + cabozantinib vs sunitinib	PFS 16.6 vs 8.3, HR 0.51; OS HR 0.70	All risk groups
CLEAR	Pembrolizumab + lenvatinib vs sunitinib	PFS 23.9 vs 9.2, **HR 0.39**; response 71%	Largest PFS/response effect

★ Putting it together

A · SMALL RENAL MASS

	Active surveillance (AS)	Ablation (Ch 143)	Partial nephrectomy (PN)	Radical nephrectomy (RN)
Best for	Frail, life expectancy <5 y; <2 cm or cystic	cT1a **≤3 cm**, comorbid; biopsy first	cT1a standard; imperative if solitary, bilateral, hereditary or CKD	Only if high complexity, no CKD/proteinuria, and new eGFR **>45**
Numbers	Growth 0.1–0.3 cm/y; treat if >3 cm or >5 mm/y	Cryo: 2 freeze–thaw cycles; RFA/MWA >60–70 °C	Trifecta: R0, no complication, warm ischaemia ≤25 min	New GFR ≈ GFR × contralateral split × 1.25
Weakness	Imaging adherence	More local recurrence; no pathology	Leak, bleeding	CKD 65% vs 20% after PN

B · RENAL LESIONS

Lesion	Imaging	Genetics / IHC	Action
Simple cyst	≤20 HU, no enhancement	—	Nothing
Complex cyst	Bosniak v2019	—	≤2 cm surveil; >4 cm excise
Angiomyolipoma (AML)	Fat ≤−20 HU	TSC2/TSC1; HMB-45, Melan-A+	Embolise if >6 cm, aneurysm >5 mm or bleeding; everolimus in TSC
Oncocytoma	Central scar	KIT+, CK7 single cells; Birt–Hogg–Dubé	Surveil or PN (chromophobe mimic)
Clear cell RCC	Hypervascular	3p/VHL loss activates HIF-2α; CA-IX	PN/RN/ablation/AS
Papillary RCC	Hypoenhancing	Trisomy 7, 17; loss of Y; MET	As RCC
Chromophobe RCC	Moderate enhancement	Multiple losses; diffuse CK7	As RCC; good prognosis
Lymphoma/metastasis	Infiltrative	—	**Biopsy; don't operate**
UTUC	Filling defect, ~46 HU, cytology+	MSI suggests Lynch	Risk-stratify, then KSM or RNU

C · UTUC: KIDNEY-SPARING (KSM) VS RADICAL NEPHROURETERECTOMY (RNU)

	KSM (endoscopic or segmental)	RNU + bladder cuff
Who	Low risk: ALL of unifocal, <2 cm, low grade, no invasion on CT urogram; or imperative (solitary, CKD, bilateral)	ANY of multifocal, >2 cm, high grade, hydronephrosis, prior cystectomy, variant
Extras	Surveillance ureteroscopy; UGN-101 mitomycin gel	≤12 wk; template node dissection; single post-op intravesical chemo
Systemic	—	Neoadjuvant platinum if post-op CrCl <50; adjuvant (POUT) ≥pT2/N+

D · OPEN VS MINIMALLY INVASIVE

- **Open (Ch 141)** is kept for IVC thrombus needing bypass, complex anatomy, prior surgery and trauma.
- **Laparoscopic/robotic (Ch 142)** is standard for organ-confined RN and most PN, with faster recovery and the same oncology.

E · METASTATIC RCC

ALGORITHM

1. **Score IMDC (6 factors, Ch 144)**: 0 is favourable, 1–2 intermediate, ≥3 poor.
2. **First line**: IO + TKI (pembro + axitinib, nivo + cabozantinib, pembro + lenvatinib) at any risk; ipi + nivo only if intermediate/poor.
3. **Cytoreductive nephrectomy**: upfront only if fit and low burden; otherwise deferred (CARMENA, SURTIME).
4. **Later lines**: switch mechanism; belzutifan after IO + TKI.